American BioCarbon® Bagasse-Based Biochar

Technical, Nutritional & Environmental Report

1 Executive Summary

American BioCarbon® (AB) Bagasse-based Biochar is a 100% organic, carbon-rich soil amendment produced from 100% sugarcane bagasse in Louisiana, USA.

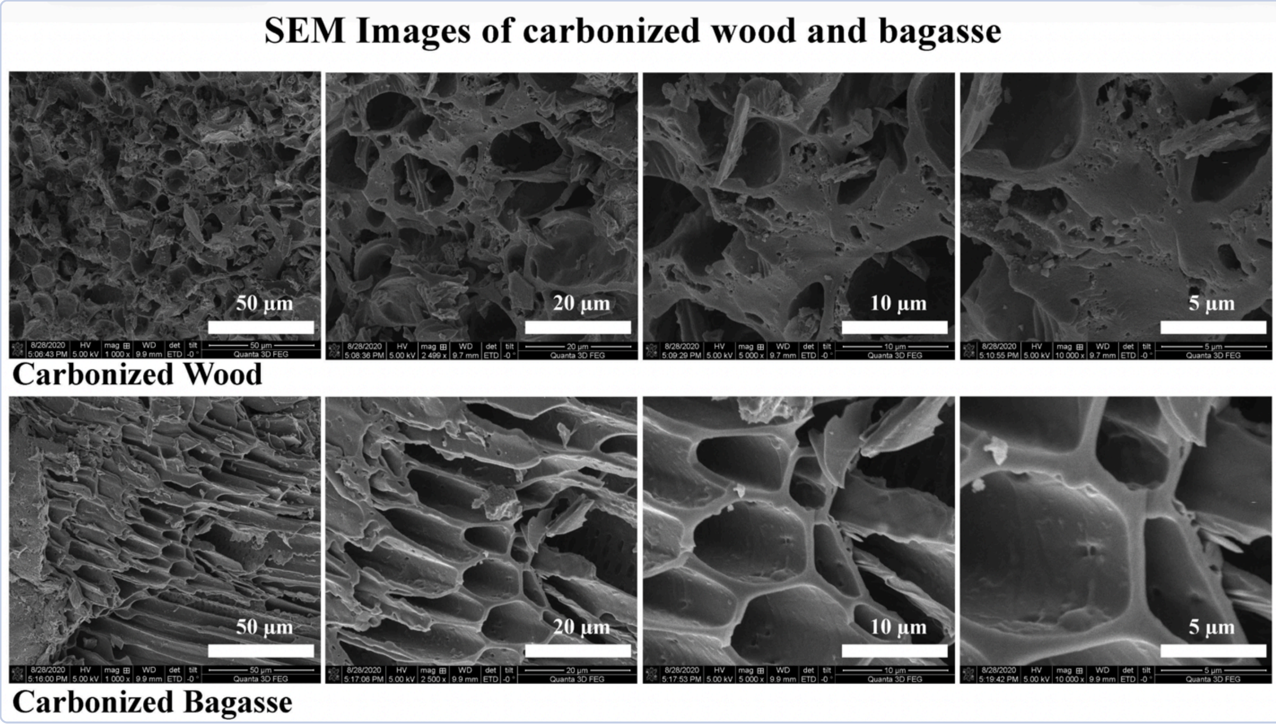
Based on 12 independent IBI laboratory analyses (attached), the product demonstrates consistent quality, stable carbon structure, measurable inherent nutrients (N-P-K), and extremely low heavy metal concentrations.

AB Bagasse-based Biochar functions simultaneously as:

- A **soil conditioner** (structure, porosity, water retention)
- A **nutrient-retention matrix**
- A **slow-release source of naturally occurring plant nutrients**
- A **long-term soil carbon input**

2 Feedstock & Sustainability

- Feedstock: **Untreated sugarcane bagasse**
- Source: Agricultural by-product of cane sugar manufacturing
- **No trees cut**
- No treated wood, biosolids, or municipal waste
- Renewable, annually replenished biomass



Structural Differences Observed by SEM

Carbonized Wood Biochar

OBSERVED STRUCTURE

- Irregular, fractured pore network
- Predominantly **collapsed vessels and disconnected pores**
- Wide variability in pore size and shape
- More brittle carbon matrix with broken cell walls

IMPLICATIONS

- Lower effective pore connectivity
- Water is held primarily in **macropores**, which drain more readily
- Reduced internal surface area available for nutrient adsorption
- Minimal inherent nutrient contribution (low ash content)

Carbonized Bagasse-Based Biochar

OBSERVED STRUCTURE

- Highly **ordered, honeycomb-like cellular architecture**
- Long, continuous channels aligned with original vascular fiber structure
- Smooth, intact pore walls with high interconnectivity
- Uniform pore dimensions at the 10-50 µm scale

IMPLICATIONS

- Significantly higher **capillary water retention**
- Stronger nutrient adsorption and retention within pore walls
- Enhanced microbial colonization due to protected pore habitats
- Higher mineral ash retention (K, P, Ca, Mg) embedded in the carbon matrix

Functional Comparison (Soil Performance)

Property	Wood Biochar	Bagasse Biochar
Pore geometry	Random, fractured	Ordered, honeycomb
Pore connectivity	Low-moderate	High
Water-holding capacity	~2× own weight	**~3-3.5× own weight**
Nutrient retention	Adsorptive only	Adsorptive + inherent nutrients
Ash / mineral content	Low	Moderate-high
Microbial habitat	Limited protection	Excellent protected niches
Performance in sandy soils	Moderate	Excellent
Performance in clay soils	Moderate	Improves porosity & drainage

Why This Matters Agronomically

- **Bagasse biochar** behaves like a **biological sponge**: it stores water and nutrients during wet periods and releases them gradually during drought or nutrient stress.
- **The continuous pore channels visible in the SEM images explain why bagasse biochar**:
  - Retains more water
  - Reduces fertilizer leaching
  - Improves nutrient use efficiency

- Wood biochar, while still beneficial for carbon sequestration, lacks the same pore regularity and nutrient-bearing ash, making it primarily a structural soil conditioner rather than a combined conditioner + nutrient contributor.

Wood-based Biochar -v- Bagasse-based Biochar

Summary Statement

SEM imaging clearly demonstrates that bagasse-derived biochar possesses a uniquely ordered, honeycomb-like pore structure with superior pore connectivity compared to wood-derived biochar. This structural difference directly translates into higher water-holding capacity, improved nutrient retention, enhanced microbial habitat, and superior agronomic performance, particularly in sandy and degraded soils.

3 Biochar Manufacturing Process

- Oxygen-limited thermal carbonization (>500 °C)
- No chemical activators, binders, acids, or bases
- Syngas generated during pyrolysis is captured and oxidized for process heat
- Solid mass reduction ~65-75%, concentrating stable carbon and mineral ash
- All production batches meet IBI H/C molar ratio < 0.7, confirming long-term carbon stability

4 Physical & Chemical Characteristics

Parameter	Typical Value
Organic Carbon	~58-65% (dry basis)
Ash Content	~25-40%
H/C Molar Ratio	0.38-0.61 (≤0.7)
pH	~7.6-9.3
Surface Area	~215-300 m^2/g
Bulk Density	~10-23 lb/ft^3
EC	<0.35 dS/m

These properties support:

- high water-holding capacity,
- nutrient adsorption,
- microbial habitat.

5 Carbon Stability & Soil Conditioning Function

The honeycomb-like porous structure typical of bagasse-derived biochar:

- Improves soil aggregation and porosity
- Increases plant-available water
- Reduces nutrient leaching
- Provides long-term habitat for beneficial microorganisms

The carbon fraction is highly recalcitrant, supporting decades- to centuries-scale soil carbon retention.

6 Nutritional Value as a Soil Amendment

Unlike many wood-based biochar, AB Bagasse-based Biochar contains measurable, naturally occurring nutrients retained from the original biomass and mineral ash.

Average Nutrient Content (Dry Basis)	
Elemental	
Nitrogen (N)	**~0.64%**
Phosphorus (P)	**~0.08%**
Potassium (K)	**~0.55%**
Secondary nutrients commonly present:	
Calcium (Ca)	~0.8-1.5%
Magnesium (Mg)	~1.0-2.0%

FERTILIZER EQUIVALENT

N - P_2O_5 - K_2O

NPK

0.6 - 0.2 - 0.7

Important:
Nutrients are **inherent, not added**, and released gradually through soil biological processes. AB Biochar contains organic fertilizer, that contribute to soil fertility and nutrient efficiency.

7 Heavy Metal & Trace Element Analysis

SUMMARY

Heavy metals in AB Bagasse-based Biochar are far below IBI, EPA Class A, and organic-use concern thresholds.

Average Heavy Metals (mg/kg, dry basis)

Element	Avg	IBI Threshold
Arsenic (As)	~0.55	13
Cadmium (Cd)	~0.31	1.4
Chromium (Cr)	~14.5	93
Cobalt (Co)	~1.5	34
Copper (Cu)	~7.4	143
Lead (Pb)	~2.7	121
Mercury (Hg)	ND	1
Nickel (Ni)	~6.1	47
Selenium (Se)	ND	2
Zinc (Zn)	~105	416

All values are one to two orders of magnitude below limits.

8 Source of Minerals & Trace Elements

Trace elements originate from:

- Naturally occurring Louisiana sugarcane field soils
- Mineral ash retained during carbonization
- Concentration via mass reduction (no addition)

9 Agronomic Performance Summary

AB Bagasse-Based Biochar:

- Improves water-holding capacity (≈3-3.5× its weight)
- Reduces fertilizer runoff and leaching
- Enhances nutrient use efficiency
- Improves performance in sandy and heavy clay soils
- Supports drought resilience

10 Regulatory & Certification Positioning

- Nonsynthetic soil amendment
- Inherent plant nutrients
- Heavy metals below concern thresholds
- IBI certified
- USDA Organic certification (pending)
- OMRI Certified
- Compatible with carbon removal and soil carbon accounting frameworks

11 Intended Use

- Soil amendment and soil conditioner
- Nutrient retention matrix
- Long-term soil carbon input
- Suitable for agriculture, horticulture, land reclamation, and soil remediation

12 Conclusion

> Based on 12 independent IBI laboratory analyses, American BioCarbon® Bagasse-Based biochar is a high-quality, nutrient-bearing biochar that delivers both physical soil improvement and inherent fertility benefits, while meeting the strictest industry and regulatory standards.

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Batch:
APR 23 A
CODE:
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Date Received: 4/2/2023
Sample ID: B-1-1-18-23-33
Lab ID. Number: 3030075-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	29.1		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	19.2		lb/cu ft	
Organic Carbon	59.1		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.61	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	36.6		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.67		% of total dry mass	Dry Combustion
pH value	8.43		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.155		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	4.7		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.5		%CaCO3	ASTM D 4373
Butane Act.	3.0		g/100g dry	ASTM D 5742-95
Surface Area Correlation	230		m2/g dry	G

All units mg/kg dry unless stated:

		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	**ND**	13 to 100	0.48	J
Cadmium	(Cd)	0.25	1.4 to 39	0.19	J
Chromium	(Cr)	5.8	93 to 1200	0.48	J
Cobalt	(Co)	1.0	34 to 100	0.48	J
Copper	(Cu)	7.2	143 to 6000	0.48	J
Lead	(Pb)	2.6	121 to 300	0.19	J
Molybdenum	(Mo)	0.81	5 to 75	0.48	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	4.3	47 to 420	0.48	J
Selenium	(Se)	**ND**	2 to 200	0.97	J
Zinc	(Zn)	127	416 to 7400	0.97	J
Boron	(B)	5.6	Declaration	4.8	TMECC
Chlorine	(Cl)	26.8	Declaration	20.0	TMECC
Sodium	(Na)	**ND**	Declaration	483	E
Iron	(Fe)	6769	Declaration	24.1	E
Manganese	(Mn)	103	Declaration	0.48	J

Particle Size Distribution

	Results	Units	Method
< 0.5mm	85.0	percent	F
0.5-1mm	10.5	percent	F
1-2mm	3.7	percent	F
2-4mm	0.8	percent	F
4-8mm	0.0	percent	F
8-16mm	0.0	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F

Basic Soil Enhancement Properties

Total (K)	4083	mg/kg	E
Total (P)	686	mg/kg	E
Ammonia (NH4-N)	9.4	mg/kg	A
Nitrate (NO3-N)	1.7	mg/kg	A
Organic (Org-N)	6696	mg/kg	Calc.
Volatile Matter	13.6	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular
G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

Analyst: Nik Zumberge

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Date Received: 5/22/2023
Sample ID: B-2-2-28-23-37
Lab ID. Number: 3050501-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	38.1		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	14.4		lb/cu ft	
Organic Carbon	61.9		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.37	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	34.7		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.62		% of total dry mass	Dry Combustion
pH value	7.92		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.306		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	6.4		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.2		%CaCO3	ASTM D 4373
Butane Act.	3.8		g/100g dry	ASTM D 5742-95
Surface Area Correlation	254		m2/g dry	G

All units mg/kg dry unless stated:		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	0.51	13 to 100	0.43	J
Cadmium	(Cd)	0.20	1.4 to 39	0.17	J
Chromium	(Cr)	16.4	93 to 1200	0.43	J
Cobalt	(Co)	1.4	34 to 100	0.43	J
Copper	(Cu)	18.9	143 to 6000	0.43	J
Lead	(Pb)	3.2	121 to 300	0.17	J
Molybdenum	(Mo)	5.2	5 to 75	0.43	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	5.4	47 to 420	0.43	J
Selenium	(Se)	**ND**	2 to 200	0.85	J
Zinc	(Zn)	78.1	416 to 7400	0.85	J
Boron	(B)	12.8	Declaration	4.3	TMECC
Chlorine	(Cl)	121	Declaration	20.0	TMECC
Sodium	(Na)	2764	Declaration	427	E
Iron	(Fe)	5174	Declaration	21.4	E
Manganese	(Mn)	107	Declaration	0.43	J

Particle Size Distribution	Results	Units	Method
< 0.5mm	96.2	percent	F
0.5-1mm	3.0	percent	F
1-2mm	0.3	percent	F
2-4mm	0.3	percent	F
4-8mm	0.2	percent	F
8-16mm	0.0	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F

Basic Soil Enhancement Properties	Results	Units	Method
Total (K)	5418	mg/kg	E
Total (P)	673	mg/kg	E
Ammonia (NH4-N)	9.0	mg/kg	A
Nitrate (NO3-N)	2.7	mg/kg	A
Organic (Org-N)	6238	mg/kg	Calc.
Volatile Matter	10.1	percent dw	D

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J EPA3050B/EPA 6020

Analyst: Nik Zumberge

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Date Received: 6/8/2023
Sample ID: B-2-3-21-23-33
Lab ID. Number: 3060184-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	51.6		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	9.8		lb/cu ft	
Organic Carbon	64.3		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.50	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	25.3		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.64		% of total dry mass	Dry Combustion
pH value	8.69		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.180		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	10.7		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.7		%CaCO3	ASTM D 4373
Butane Act.	4.0		g/100g dry	ASTM D 5742-95
Surface Area Correlation	259		m2/g dry	G

All units mg/kg dry unless stated:		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	0.45	13 to 100	0.44	J
Cadmium	(Cd)	0.28	1.4 to 39	0.17	J
Chromium	(Cr)	16.0	93 to 1200	0.44	J
Cobalt	(Co)	1.5	34 to 100	0.44	J
Copper	(Cu)	8.2	143 to 6000	0.44	J
Lead	(Pb)	2.8	121 to 300	0.17	J
Molybdenum	(Mo)	1.0	5 to 75	0.44	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	7.9	47 to 420	0.44	J
Selenium	(Se)	**ND**	2 to 200	0.87	J
Zinc	(Zn)	57.5	416 to 7400	0.87	J
Boron	(B)	15.1	Declaration	4.4	TMECC
Chlorine	(Cl)	184	Declaration	20.0	TMECC
Sodium	(Na)	831	Declaration	437	E
Iron	(Fe)	5727	Declaration	21.9	E
Manganese	(Mn)	244	Declaration	0.44	J

Particle Size Distribution	Results	Units	Method
< 0.5mm	92.4	percent	F
0.5-1mm	4.8	percent	F
1-2mm	0.4	percent	F
2-4mm	0.5	percent	F
4-8mm	0.1	percent	F
8-16mm	1.7	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F
Basic Soil Enhancement Properties			
Total (K)	5338	mg/kg	E
Total (P)	836	mg/kg	E
Ammonia (NH4-N)	3.6	mg/kg	A
Nitrate (NO3-N)	2.0	mg/kg	A
Organic (Org-N)	6402	mg/kg	Calc.
Volatile Matter	31.3	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular
G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

Analyst: Nik Zumberge

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Date Received: 8/8/2023
Sample ID: B-1-6-6-23
Lab ID. Number: 3080165-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	43.0		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	10.5		lb/cu ft	
Organic Carbon	62.9		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.51	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	27.8		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.65		% of total dry mass	Dry Combustion
pH value	7.73		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.077		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	6.0		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.4		%CaCO3	ASTM D 4373
Butane Act.	4.1		g/100g dry	ASTM D 5742-95
Surface Area Correlation	264		m2/g dry	G

All units mg/kg dry unless stated:		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	0.60	13 to 100	0.43	J
Cadmium	(Cd)	0.31	1.4 to 39	0.17	J
Chromium	(Cr)	18.3	93 to 1200	0.43	J
Cobalt	(Co)	1.3	34 to 100	0.43	J
Copper	(Cu)	6.9	143 to 6000	0.43	J
Lead	(Pb)	2.6	121 to 300	0.17	J
Molybdenum	(Mo)	0.64	5 to 75	0.43	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	8.7	47 to 420	0.43	J
Selenium	(Se)	**ND**	2 to 200	0.85	J
Zinc	(Zn)	47.5	416 to 7400	0.85	J
Boron	(B)	**ND**	Declaration	4.3	TMECC
Chlorine	(Cl)	211	Declaration	20.0	TMECC
Sodium	(Na)	**ND**	Declaration	427	E
Iron	(Fe)	4594	Declaration	21.3	E
Manganese	(Mn)	98.6	Declaration	0.43	J

Particle Size Distribution	Results	Units	Method
< 0.5mm	77.1	percent	F
0.5-1mm	18.8	percent	F
1-2mm	2.0	percent	F
2-4mm	0.2	percent	F
4-8mm	0.7	percent	F
8-16mm	1.3	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F
Basic Soil Enhancement Properties			
Total (K)	5342	mg/kg	E
Total (P)	684	mg/kg	E
Ammonia (NH4-N)	9.7	mg/kg	A
Nitrate (NO3-N)	5.1	mg/kg	A
Organic (Org-N)	6478	mg/kg	Calc.
Volatile Matter	15.9	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
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G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

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Date Received: 10/12/2023
Sample ID: Week 40, 2023 B-3-10-4-23
Lab ID. Number: 3100311-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	30.4		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	10.1		lb/cu ft	
Organic Carbon	69.9		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.40	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	23.8		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.64		% of total dry mass	Dry Combustion
pH value	8.83		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.154		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	7.7		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	6.5		%CaCO3	ASTM D 4373
Butane Act.	4.6		g/100g dry	ASTM D 5742-95
Surface Area Correlation	278		m2/g dry	G

All units mg/kg dry unless stated:		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	**ND**	13 to 100	0.50	J
Cadmium	(Cd)	0.50	1.4 to 39	0.20	J
Chromium	(Cr)	30.7	93 to 1200	0.50	J
Cobalt	(Co)	2.0	34 to 100	0.50	J
Copper	(Cu)	7.2	143 to 6000	0.50	J
Lead	(Pb)	2.6	121 to 300	0.20	J
Molybdenum	(Mo)	0.74	5 to 75	0.50	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	13.6	47 to 420	0.50	J
Selenium	(Se)	**ND**	2 to 200	1.00	J
Zinc	(Zn)	132	416 to 7400	1.00	J
Boron	(B)	42.1	Declaration	5.0	TMECC
Chlorine	(Cl)	106	Declaration	20.0	TMECC
Sodium	(Na)	1114	Declaration	499	E
Iron	(Fe)	7649	Declaration	24.9	E
Manganese	(Mn)	129	Declaration	0.50	J

Particle Size Distribution	Results	Units	Method
< 0.5mm	57.6	percent	F
0.5-1mm	28.4	percent	F
1-2mm	13.7	percent	F
2-4mm	0.4	percent	F
4-8mm	0.0	percent	F
8-16mm	0.0	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F
Basic Soil Enhancement Properties			
Total (K)	5109	mg/kg	E
Total (P)	923	mg/kg	E
Ammonia (NH4-N)	1.0	mg/kg	A
Nitrate (NO3-N)	1.0	mg/kg	A
Organic (Org-N)	6415	mg/kg	Calc.
Volatile Matter	9.1	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular
G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

Analyst: Nik Zumberge

Control Laboratories

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Account No:
11499
Batch:
Oct 23 B
CODE:
BioChar IBI

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Date Received:	10/12/2023
Sample ID:	Week 39, 2023 B-4-9-27-23
Lab ID. Number:	3100312-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	30.7		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	10.1		lb/cu ft	
Organic Carbon	70.2		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.38	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	24.5		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.66		% of total dry mass	Dry Combustion
pH value	8.58		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.128		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	7.2		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.4		%CaCO3	ASTM D 4373
Butane Act.	4.3		g/100g dry	ASTM D 5742-95
Surface Area Correlation	270		m2/g dry	G

All units mg/kg dry unless stated:		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	**ND**	13 to 100	0.49	J
Cadmium	(Cd)	0.53	1.4 to 39	0.20	J
Chromium	(Cr)	18.5	93 to 1200	0.49	J
Cobalt	(Co)	1.7	34 to 100	0.49	J
Copper	(Cu)	5.6	143 to 6000	0.49	J
Lead	(Pb)	2.9	121 to 300	0.20	J
Molybdenum	(Mo)	**ND**	5 to 75	0.49	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	6.7	47 to 420	0.49	J
Selenium	(Se)	**ND**	2 to 200	0.99	J
Zinc	(Zn)	131	416 to 7400	0.99	J
Boron	(B)	59.4	Declaration	4.9	TMECC
Chlorine	(Cl)	69	Declaration	20.0	TMECC
Sodium	(Na)	564	Declaration	493	E
Iron	(Fe)	6375	Declaration	24.7	E
Manganese	(Mn)	116	Declaration	0.49	J

Particle Size Distribution	Results	Units	Method
< 0.5mm	70.2	percent	F
0.5-1mm	26.0	percent	F
1-2mm	3.4	percent	F
2-4mm	0.4	percent	F
4-8mm	0.0	percent	F
8-16mm	0.0	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F
Basic Soil Enhancement Properties			
Total (K)	5154	mg/kg	E
Total (P)	902	mg/kg	E
Ammonia (NH4-N)	1.6	mg/kg	A
Nitrate (NO3-N)	2.0	mg/kg	A
Organic (Org-N)	6571	mg/kg	Calc.
Volatile Matter	13.3	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular
G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

Analyst: Nik Zumberge

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Batch:
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CODE:
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Date Received: 11/22/2023
Sample ID: B-11-14-23-2
Lab ID. Number: 3110472-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	32.4		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	14.9		lb/cu ft	
Organic Carbon	62.0		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.58	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	33.9		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.72		% of total dry mass	Dry Combustion
pH value	9.18		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.319		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	8.9		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	1.6		%CaCO3	ASTM D 4373
Butane Act.	5.2		g/100g dry	ASTM D 5742-95
Surface Area Correlation	298		m2/g dry	G

All units mg/kg dry unless stated:

		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	0.36	13 to 100	0.33	J
Cadmium	(Cd)	0.24	1.4 to 39	0.13	J
Chromium	(Cr)	7.5	93 to 1200	0.33	J
Cobalt	(Co)	1.2	34 to 100	0.33	J
Copper	(Cu)	8.8	143 to 6000	0.33	J
Lead	(Pb)	2.6	121 to 300	0.13	J
Molybdenum	(Mo)	0.58	5 to 75	0.33	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	3.4	47 to 420	0.33	J
Selenium	(Se)	**ND**	2 to 200	0.65	J
Zinc	(Zn)	54.7	416 to 7400	0.65	J
Boron	(B)	10.7	Declaration	3.3	TMECC
Chlorine	(Cl)	152	Declaration	20.0	TMECC
Sodium	(Na)	342	Declaration	327	E
Iron	(Fe)	5163	Declaration	16.3	E
Manganese	(Mn)	96.6	Declaration	0.33	J

Particle Size Distribution

	Results	Units	Method
< 0.5mm	84.9	percent	F
0.5-1mm	12.6	percent	F
1-2mm	2.1	percent	F
2-4mm	0.3	percent	F
4-8mm	0.0	percent	F
8-16mm	0.0	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F

Basic Soil Enhancement Properties

Total (K)	8438	mg/kg	E
Total (P)	899	mg/kg	E
Ammonia (NH4-N)	10.2	mg/kg	A
Nitrate (NO3-N)	2.8	mg/kg	A
Organic (Org-N)	7182	mg/kg	Calc.
Volatile Matter	10.0	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular
G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

Analyst: Nik Zumberge

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Batch:
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CODE:
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Date Received:	12/27/2024
Sample ID:	B-12-7-23-1
Lab ID. Number:	3120549-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	24.4		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	22.9		lb/cu ft	
Organic Carbon	52.1		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.44	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	40.3		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.46		% of total dry mass	Dry Combustion
pH value	9.27		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.221		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	11.4		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.7		%CaCO3	ASTM D 4373
Butane Act.	5.0		g/100g dry	ASTM D 5742-95
Surface Area Correlation	292		m2/g dry	G

All units mg/kg dry unless stated:		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	0.73	13 to 100	0.36	J
Cadmium	(Cd)	0.30	1.4 to 39	0.14	J
Chromium	(Cr)	12.1	93 to 1200	0.36	J
Cobalt	(Co)	1.7	34 to 100	0.36	J
Copper	(Cu)	9.5	143 to 6000	0.36	J
Lead	(Pb)	3.4	121 to 300	0.14	J
Molybdenum	(Mo)	0.76	5 to 75	0.36	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	5.7	47 to 420	0.36	J
Selenium	(Se)	**ND**	2 to 200	0.71	J
Zinc	(Zn)	75.7	416 to 7400	0.71	J
Boron	(B)	9.3	Declaration	3.6	TMECC
Chlorine	(Cl)	100	Declaration	20.0	TMECC
Sodium	(Na)	**ND**	Declaration	357	E
Iron	(Fe)	6669	Declaration	17.9	E
Manganese	(Mn)	130	Declaration	0.36	J

Particle Size Distribution	Results	Units	Method
< 0.5mm	90.5	percent	F
0.5-1mm	6.4	percent	F
1-2mm	2.5	percent	F
2-4mm	0.5	percent	F
4-8mm	0.0	percent	F
8-16mm	0.0	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F

Basic Soil Enhancement Properties			
Total (K)	6643	mg/kg	E
Total (P)	990	mg/kg	E
Ammonia (NH4-N)	3.9	mg/kg	A
Nitrate (NO3-N)	1.2	mg/kg	A
Organic (Org-N)	4603	mg/kg	Calc.
Volatile Matter	17.2	percent dw	D

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Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular
G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

Analyst: Nik Zumberge

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Account No: 11499
Batch: MAR 24 E
CODE: BioChar IBI

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Date Received: 3/21/2024
Sample ID: B-1-3-6-24
Lab ID. Number: 4030435-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	28.3		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	14.6		lb/cu ft	
Organic Carbon	63.5		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.55	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	29.5		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.69		% of total dry mass	Dry Combustion
pH value	8.52		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.180		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	8.2		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	1.9		%CaCO3	ASTM D 4373
Butane Act.	2.9		g/100g dry	ASTM D 5742-95
Surface Area Correlation	225		m2/g dry	G

All units mg/kg dry unless stated:		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	0.59	13 to 100	0.48	J
Cadmium	(Cd)	0.33	1.4 to 39	0.19	J
Chromium	(Cr)	7.9	93 to 1200	0.48	J
Cobalt	(Co)	1.5	34 to 100	0.48	J
Copper	(Cu)	6.2	143 to 6000	0.48	J
Lead	(Pb)	2.7	121 to 300	0.19	J
Molybdenum	(Mo)	0.7	5 to 75	0.48	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	4.2	47 to 420	0.48	J
Selenium	(Se)	**ND**	2 to 200	0.96	J
Zinc	(Zn)	170	416 to 7400	0.96	J
Boron	(B)	6.4	Declaration	4.8	TMECC
Chlorine	(Cl)	116	Declaration	20.0	TMECC
Sodium	(Na)	**ND**	Declaration	482	E
Iron	(Fe)	7787	Declaration	24.1	E
Manganese	(Mn)	99.0	Declaration	0.48	J

Particle Size Distribution	Results	Units	Method
< 0.5mm	77.8	percent	F
0.5-1mm	17.7	percent	F
1-2mm	4.2	percent	F
2-4mm	0.4	percent	F
4-8mm	0.0	percent	F
8-16mm	0.0	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F
Basic Soil Enhancement Properties			
Total (K)	5330	mg/kg	E
Total (P)	869	mg/kg	E
Ammonia (NH4-N)	99.0	mg/kg	A
Nitrate (NO3-N)	2.8	mg/kg	A
Organic (Org-N)	6777	mg/kg	Calc.
Volatile Matter	16.9	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular
G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

Analyst: Nik Zumberge

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Batch:
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CODE:
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Date Received: 4/4/2024
Sample ID: B-1-2-8-24
Lab ID. Number: 4040126-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	27.6		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	13.8		lb/cu ft	
Organic Carbon	62.0		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.52	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	32.7		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.71		% of total dry mass	Dry Combustion
pH value	8.30		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.174		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	8.1		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.2		%CaCO3	ASTM D 4373
Butane Act.	2.6		g/100g dry	ASTM D 5742-95
Surface Area Correlation	217		m2/g dry	G

All units mg/kg dry unless stated:

		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	**ND**	13 to 100	0.39	J
Cadmium	(Cd)	0.19	1.4 to 39	0.16	J
Chromium	(Cr)	4.3	93 to 1200	0.39	J
Cobalt	(Co)	0.84	34 to 100	0.39	J
Copper	(Cu)	4.1	143 to 6000	0.39	J
Lead	(Pb)	1.6	121 to 300	0.16	J
Molybdenum	(Mo)	**ND**	5 to 75	0.39	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	2.4	47 to 420	0.39	J
Selenium	(Se)	**ND**	2 to 200	0.78	J
Zinc	(Zn)	94.1	416 to 7400	0.78	J
Boron	(B)	4.3	Declaration	3.9	TMECC
Chlorine	(Cl)	434	Declaration	20.0	TMECC
Sodium	(Na)	**ND**	Declaration	388	E
Iron	(Fe)	4109	Declaration	19.4	E
Manganese	(Mn)	54.2	Declaration	0.39	J

Particle Size Distribution

	Results	Units	Method
< 0.5mm	75.4	percent	F
0.5-1mm	18.9	percent	F
1-2mm	5.3	percent	F
2-4mm	0.4	percent	F
4-8mm	0.0	percent	F
8-16mm	0.0	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F

Basic Soil Enhancement Properties

	Results	Units	Method
Total (K)	2974	mg/kg	E
Total (P)	431	mg/kg	E
Ammonia (NH4-N)	46.7	mg/kg	A
Nitrate (NO3-N)	2.8	mg/kg	A
Organic (Org-N)	7099	mg/kg	Calc.
Volatile Matter	23.4	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular
G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

Analyst: Nik Zumberge

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Batch:
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CODE:
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Date Received: 4/4/2024
Sample ID: B-1-1-19-24
Lab ID. Number: 4040127-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	27.1		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	14.1		lb/cu ft	
Organic Carbon	59.6		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.55	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	28.9		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.70		% of total dry mass	Dry Combustion
pH value	8.51		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.152		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	13.8		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.0		%CaCO3	ASTM D 4373
Butane Act.	2.5		g/100g dry	ASTM D 5742-95
Surface Area Correlation	213		m2/g dry	G

All units mg/kg dry unless stated:		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	0.53	13 to 100	0.42	J
Cadmium	(Cd)	0.30	1.4 to 39	0.17	J
Chromium	(Cr)	7.5	93 to 1200	0.42	J
Cobalt	(Co)	1.42	34 to 100	0.42	J
Copper	(Cu)	6.3	143 to 6000	0.42	J
Lead	(Pb)	2.7	121 to 300	0.17	J
Molybdenum	(Mo)	0.69	5 to 75	0.42	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	4.3	47 to 420	0.42	J
Selenium	(Se)	**ND**	2 to 200	0.84	J
Zinc	(Zn)	132	416 to 7400	0.84	J
Boron	(B)	6.5	Declaration	4.2	TMECC
Chlorine	(Cl)	100	Declaration	20.0	TMECC
Sodium	(Na)	**ND**	Declaration	420	E
Iron	(Fe)	6973	Declaration	21.0	E
Manganese	(Mn)	98.3	Declaration	0.42	J

Particle Size Distribution	Results	Units	Method
< 0.5mm	70.5	percent	F
0.5-1mm	20.3	percent	F
1-2mm	8.6	percent	F
2-4mm	0.5	percent	F
4-8mm	0.0	percent	F
8-16mm	0.0	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F
Basic Soil Enhancement Properties			
Total (K)	5272	mg/kg	E
Total (P)	806	mg/kg	E
Ammonia (NH4-N)	8.0	mg/kg	A
Nitrate (NO3-N)	1.7	mg/kg	A
Organic (Org-N)	6956	mg/kg	Calc.
Volatile Matter	22.5	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular
G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

Analyst: Nik Zumberge

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Account No:
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Batch:
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CODE:
BioChar IBI

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Date Received:	4/25/2024
Sample ID:	B-4-4-15-24
Lab ID. Number:	4040648-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	39.9		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	16.3		lb/cu ft	
Organic Carbon	59.9		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.46	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	32.8		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.61		% of total dry mass	Dry Combustion
pH value	7.59		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.141		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	6.7		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.6		%CaCO3	ASTM D 4373
Butane Act.	3.2		g/100g dry	ASTM D 5742-95
Surface Area Correlation	235		m2/g dry	G

All units mg/kg dry unless stated:		Results	Range of Max. Levels	Reporting Limit (ppm)	Method
Arsenic	(As)	0.56	13 to 100	0.43	J
Cadmium	(Cd)	0.36	1.4 to 39	0.17	J
Chromium	(Cr)	10.8	93 to 1200	0.43	J
Cobalt	(Co)	1.6	34 to 100	0.43	J
Copper	(Cu)	5.8	143 to 6000	0.43	J
Lead	(Pb)	2.6	121 to 300	0.17	J
Molybdenum	(Mo)	0.87	5 to 75	0.43	J
Mercury	(Hg)	**ND**	1 to 17	0.001	EPA 7471
Nickel	(Ni)	6.0	47 to 420	0.43	J
Selenium	(Se)	**ND**	2 to 200	0.85	J
Zinc	(Zn)	134	416 to 7400	0.85	J
Boron	(B)	8.6	Declaration	4.3	TMECC
Chlorine	(Cl)	58.6	Declaration	20.0	TMECC
Sodium	(Na)	512	Declaration	427	E
Iron	(Fe)	7866	Declaration	21.4	E
Manganese	(Mn)	110	Declaration	0.43	J

Particle Size Distribution	Results	Units	Method
< 0.5mm	72.9	percent	F
0.5-1mm	16.3	percent	F
1-2mm	8.8	percent	F
2-4mm	1.4	percent	F
4-8mm	0.6	percent	F
8-16mm	0.0	percent	F
16-25mm	0.0	percent	F
25-50mm	0.0	percent	F
>50mm	0.0	percent	F

Basic Soil Enhancement Properties	Results	Units	Method
Total (K)	5226	mg/kg	E
Total (P)	862	mg/kg	E
Ammonia (NH4-N)	2.5	mg/kg	A
Nitrate (NO3-N)	1.3	mg/kg	A
Organic (Org-N)	6097	mg/kg	Calc.
Volatile Matter	15.2	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular
G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

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